UNITED STATES INTERNATIONAL UNIVERSITY – AFRICA

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PREDICTING MATERNAL HEALTH RISK AMONG PREGNANT WOMEN IN KENYA USING MACHINE LEARNING AND BAYESIAN INFERENCE ON THE KENYA DEMOGRAPHIC AND HEALTH SURVEY 2022

A Research Proposal Submitted in Partial Fulfilment of the Requirements for the Bachelor of Science in Data Science and Analytics

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STUDENT'S DECLARATION

I, **Selmah Tzindori Mise**, declare that this research proposal titled *"Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference on the Kenya Demographic and Health Survey 2022"* is my own original work and has not been submitted for any degree or examination at any other university. All sources of information used in this proposal have been duly acknowledged through appropriate citations and references.

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ABSTRACT

Maternal mortality remains a critical public health challenge in Kenya, with an estimated 355 deaths per 100,000 live births, far exceeding the Sustainable Development Goal 3.1 target of 70 per 100,000. The majority of these deaths are preventable with timely identification of high-risk pregnancies and appropriate referral. Despite the deployment of over 100,000 Community Health Workers (CHWs) across Kenya's 47 counties, no data-driven, individual-level maternal risk prediction tool exists to support their clinical decision-making in the field.

This study proposes to develop a machine learning-based maternal health risk prediction system using the Kenya Demographic and Health Survey 2022 (KDHS 2022) Individual Recode dataset. The system will classify individual pregnant women as Low, Moderate, or High risk based on 13 demographic, socioeconomic, and reproductive health variables. A four-tier machine learning pipeline, incorporating baseline models (Logistic Regression, Decision Tree), classical ensemble methods (Random Forest, XGBoost, CatBoost), novel deep learning architectures for tabular data (TabNet, TabTransformer), and a stacking ensemble, will be developed and comparatively evaluated. Bayesian inference using PyMC will augment each prediction with calibrated posterior probability scores and 95% credible intervals. SHAP (SHapley Additive exPlanations) values will provide model-level and individual-level explainability. The end product is a Streamlit web application with AI API integration that generates plain-language risk explanations, care recommendations, and referral letters for CHWs in English and Swahili.

The study addresses a confirmed gap in the literature: no existing study has applied novel tabular deep learning architectures with Bayesian uncertainty quantification to individual-level KDHS data for CHW deployment in Kenya. The proposed system has potential to directly contribute to SDG 3.1 and Kenya's Universal Health Coverage agenda by placing data-driven decision support in the hands of frontline community health workers.

Keywords: Maternal health risk, Machine learning, Bayesian inference, Kenya, KDHS 2022, TabNet, TabTransformer, Community Health Workers, Explainable AI, SHAP

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Selmah Tzindori Mise

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Contents

Student's Declaration **1**

Copyright **2**

Abstract **3**

Acknowledgements **4**

1 CHAPTER ONE: INTRODUCTION **11**

CHAPTER ONE: INTRODUCTION **11**

1.1 Background to the Problem 11

Background to the Problem 11

1.2 Statement of the Problem 12

Statement of the Problem 12

1.3 Objectives of the Study 13

Objectives of the Study 13

1.3.1 General Objective 13

General Objective 13

1.3.2 Specific Objectives 13

Specific Objectives 13

1.4 Research Questions 13

Research Questions 13

1.5 Purpose of the Study 14

Purpose of the Study 14

1.6 Justification of the Study 14

Justification of the Study 14

1.6.1 Research Gap 14

Research Gap 14

1.6.2 Why the Recommended Approach 15

Why the Recommended Approach 15

1.6.3 Significance of the Study 15

Significance of the Study 15

1.7 Limitations of the Study 16

Limitations of the Study 16

1.8 Definition of Terms . . . 17
Definition of Terms . . . 17
1.9 Chapter Summary . . . 18
Chapter Summary . . . 18
2 CHAPTER TWO: LITERATURE REVIEW . . . 20
CHAPTER TWO: LITERATURE REVIEW . . . 20
2.1 Introduction . . . 20
Introduction . . . 20
2.2 General Literature Review . . . 20
General Literature Review . . . 20
2.2.1 Maternal Mortality — Global Burden and Kenyan Context . . . 20
Maternal Mortality — Global Burden and Kenyan Context . . . 20
2.2.2 Kenya's Community Health System and Digital Health Landscape 21
Kenya's Community Health System and Digital Health Landscape . . . 21
2.2.3 Machine Learning in Healthcare — Evolution and Novel Architectures 21
Machine Learning in Healthcare — Evolution and Novel Architectures . . 21
2.2.4 Explainable Artificial Intelligence in Clinical Settings . . . 22
Explainable Artificial Intelligence in Clinical Settings . . . 22
2.3 Empirical Literature Review . . . 23
Empirical Literature Review . . . 23
2.3.1 Studies Related to Objective 1 — ML Classification for Maternal Risk . . . 23
Studies Related to Objective 1 — ML Classification for Maternal Risk . . 23
2.3.2 Studies Related to Objective 2 — Bayesian Inference for Risk Quantification . . . 24
Studies Related to Objective 2 — Bayesian Inference for Risk Quantification 24
2.3.3 Studies Related to Objective 3 — AI-Powered CHW Decision Support 24
Studies Related to Objective 3 — AI-Powered CHW Decision Support . . 24
2.4 Research Gaps . . . 25
Research Gaps . . . 25
2.5 Chapter Summary . . . 26

Chapter Summary 26
3 CHAPTER THREE: RESEARCH METHODOLOGY 27
CHAPTER THREE: RESEARCH METHODOLOGY 27
3.1 Introduction 27
Introduction 27
3.2 Design and Experimental Methodology 27
Design and Experimental Methodology 27
3.3 Research Design 27
Research Design 27
3.4 Mathematical Foundations and Model Equations 28
Mathematical Foundations and Model Equations 28
3.4.1 Target Variable Engineering — Composite Risk Score 28
Target Variable Engineering — Composite Risk Score 28
3.4.2 Data Preprocessing — SMOTE 29
Data Preprocessing — SMOTE 29
3.4.3 Tier 1 — Logistic Regression (Baseline) 29
Tier 1 — Logistic Regression (Baseline) 29
3.4.4 Tier 2 — Random Forest 29
Tier 2 — Random Forest 29
3.4.5 Tier 2 — XGBoost (Extreme Gradient Boosting) 30
Tier 2 — XGBoost (Extreme Gradient Boosting) 30
3.4.6 Tier 3 — TabNet (Novel Architecture) 31
Tier 3 — TabNet (Novel Architecture) 31
3.4.7 Tier 3 — TabTransformer (Novel Architecture) 32
Tier 3 — TabTransformer (Novel Architecture) 32
3.4.8 Tier 4 — Stacking Ensemble 33
Tier 4 — Stacking Ensemble 33
3.4.9 Bayesian Inference — Posterior Distribution 33
Bayesian Inference — Posterior Distribution 33
3.4.10 SHAP — Feature Attribution 34
SHAP — Feature Attribution 34
3.4.11 Evaluation Metrics 35

Evaluation Metrics . . . 35
3.5 Data Collection Methods . . . 36
Data Collection Methods . . . 36
3.5.1 Primary Data Source . . . 36
Primary Data Source . . . 36
3.5.2 Variable Selection and Dataset Characteristics . . . 36
Variable Selection and Dataset Characteristics . . . 36
3.6 Instrumentation . . . 37
Instrumentation . . . 37
3.7 Method of Data Analysis . . . 38
Method of Data Analysis . . . 38
3.7.1 Exploratory Data Analysis . . . 38
Exploratory Data Analysis . . . 38
3.7.2 Preprocessing Pipeline . . . 39
Preprocessing Pipeline . . . 39
3.7.3 Model Training and Hyperparameter Optimisation . . . 39
Model Training and Hyperparameter Optimisation . . . 39
3.7.4 Feature Importance Analysis . . . 39
Feature Importance Analysis . . . 39
3.8 Validity and Evaluation . . . 39
Validity and Evaluation . . . 39
3.9 Ethical Considerations . . . 40
Ethical Considerations . . . 40
3.10 Chapter Summary . . . 40
Chapter Summary . . . 40
References . . . 41
APPENDICES . . . 44
Appendix A: Introduction Letter . . . 44
Appendix B: Budget . . . 46
Appendix C: Research Workplan and Timeline . . . 47
Appendix D: Data Collection Sheet / Variable Codebook . . . 49

Appendix E: Experimental Protocols 50
Protocol 1 — Model Training 50
Protocol 2 — Bayesian Inference 50
Protocol 3 — Application Validation 50
Appendix F: Dataset Description 51
Appendix G: Model and Algorithm Specifications 52
Appendix H: Informed Consent / Assent Form 53
Appendix I: Debrief Form 55
Appendix J: PI and Research Team Confidentiality Agreement 57
Appendix K: DHS Program Data Authorization Letter 59

List of Tables

2 Variables Selected for the Study 36
3 Software Libraries and Tools Used 37
4 Research Budget 46
5 Research Workplan — Summer & Fall Semester 2026 47
6 Model Hyperparameter Specifications 52

1 CHAPTER ONE: INTRODUCTION

1.1 Background to the Problem

Maternal mortality is one of the most persistent indicators of health system inequity globally. The World Health Organization (2023) defines maternal death as the death of a woman while pregnant or within 42 days of termination of pregnancy from causes related to or aggravated by the pregnancy or its management. Globally, an estimated 287,000 women died from maternal causes in 2020, with Sub-Saharan Africa accounting for approximately 70% of all maternal deaths despite representing a far smaller proportion of global births (WHO, 2023). This disproportionate burden reflects systemic failures in healthcare access, health worker capacity, and, critically, the absence of tools that enable proactive risk identification at the community level.

In Kenya, the maternal mortality ratio stands at approximately 355 deaths per 100,000 live births, according to the Kenya Demographic and Health Survey 2022 (KNBS & ICF, 2023). This figure is more than five times the Sustainable Development Goal 3.1 target of 70 deaths per 100,000 live births by 2030, and represents a slow rate of progress relative to regional peers. The leading direct causes of maternal death in Kenya include obstetric haemorrhage, hypertensive disorders of pregnancy (particularly eclampsia and pre-eclampsia), and puerperal sepsis, all of which are clinically preventable with timely identification and intervention (Say et al., 2014). Indirect causes, including severe anaemia and HIV-related complications, compound risk further, particularly among women in lower wealth quintiles and rural counties.

The Kenyan government has invested significantly in the community health system as the primary vehicle for maternal health service delivery. Over 100,000 Community Health Promoters (CHPs) are deployed across the country, trained to conduct household visits, provide antenatal care education, facilitate skilled birth attendance referrals, and report on health outcomes through the electronic Community Health Information System (eCHIS). The 2023 Social Health Insurance Act, Digital Health Act, and Primary Health Care Act collectively signal strong policy commitment to digitally enabled universal health coverage (UHC). Yet, despite this infrastructure, CHPs continue to assess maternal risk using static, paper-based checklists that do not account for the probabilistic interaction of multiple risk factors, leaving the identification of high-risk women to

clinical intuition and protocol adherence rather than data-driven evidence.

Recent advances in machine learning, deep learning for tabular data, Bayesian inference, and explainable AI (XAI) have created new possibilities for community-level health risk prediction. Novel architectures such as TabNet (Arik & Pfister, 2021) and TabTransformer (Huang et al., 2020) have demonstrated superior performance over classical ensemble methods on structured health survey data. Bayesian inference frameworks such as PyMC enable the generation of probabilistic risk estimates with quantified uncertainty, a critical capability for clinical decision support. The KDHS 2022, a nationally representative survey of 17,414 Kenyan women, provides a rich, publicly available data source that has not yet been applied to individual-level maternal risk prediction. This study proposes to leverage these methodological advances and this data resource to fill a confirmed gap in Kenyan health informatics.

1.2 Statement of the Problem

Kenya's 100,000+ Community Health Workers visit pregnant women across 47 counties daily, yet have no access to a data-driven, individual-level tool that can identify which specific women are at high risk of maternal complications before those complications occur. Existing maternal health risk assessment in Kenya relies on paper-based protocols and static clinical guidelines that treat all women with the same number of risk flags as equally at risk, without quantifying the probability or uncertainty of that risk. The result is systematic misallocation of CHW attention, missed referrals for genuinely high-risk women, and preventable maternal deaths.

Review of the existing literature confirms that while machine learning has been applied to maternal health prediction globally, no study has: (1) developed an individual-level maternal risk classification model specifically calibrated for Kenya using KDHS 2022 data; (2) incorporated novel tabular deep learning architectures (TabNet, TabTransformer) into the maternal health prediction pipeline; (3) applied Bayesian inference to generate calibrated probabilistic risk scores with credible intervals on KDHS data; or (4) integrated an AI API explanation layer to generate plain-language CHW guidance and referral letters in English and Swahili. This fourfold gap defines the problem this study addresses.

> *In summary: there is currently no deployed, individual-level, probabilistic, explainable maternal health risk scoring tool for Community Health Workers*

in Kenya, calibrated to the national demographic and health profile. This study proposes to build one.

1.3 Objectives of the Study

1.3.1 General Objective

To develop an individual-level, machine learning-based maternal health risk prediction system that classifies pregnant women in Kenya as Low, Moderate, or High risk using the KDHS 2022 dataset, with Bayesian uncertainty quantification, SHAP-based explainability, and an AI API-powered Community Health Worker decision support interface deployable at the community level.

1.3.2 Specific Objectives

- **SO1:** To build and comparatively evaluate machine learning classification models, including novel tabular deep learning architectures (TabNet and TabTransformer) alongside classical ensemble methods, for predicting individual maternal health risk using 13 variables from the KDHS 2022 dataset.
- **SO2:** To integrate Bayesian inference using PyMC to generate calibrated posterior probability distributions and 95% credible intervals for each individual maternal risk prediction, enabling uncertainty-aware clinical decision-making by Community Health Workers.
- **SO3:** To design and deploy an AI API-integrated Streamlit web application that translates model outputs, risk classifications, SHAP feature attributions, and Bayesian probability scores, into plain-language explanations, care recommendations, and referral letters for CHWs in English and Swahili.

1.4 Research Questions

- **RQ1:** Which machine learning architecture, including novel models such as TabNet and TabTransformer, achieves the highest predictive accuracy and sensitivity for the High Risk maternal class on the KDHS 2022 Individual Recode dataset?

- **RQ2:** How do Bayesian posterior probability distributions and 95% credible intervals generated using PyMC improve the actionability and reliability of individual maternal risk predictions compared to point-estimate machine learning outputs?
- **RQ3:** To what extent does the AI API explanation layer improve the interpretability and clinical usability of the maternal risk prediction tool for non-technical Community Health Workers in Kenya?

1.5 Purpose of the Study

The primary purpose of this study is to develop a deployable, evidence-based maternal health risk prediction tool that places data-driven decision support in the hands of Kenya's Community Health Workers. The study aims to demonstrate that the rich demographic and health survey data already collected by KNBS and ICF International can be operationalised without additional primary data collection to generate individual-level, probabilistic risk scores that are interpretable by non-technical frontline health workers. The secondary purpose is to establish a methodological contribution to the field of health informatics by evaluating novel tabular deep learning architectures in the maternal health prediction context and demonstrating the practical utility of Bayesian uncertainty quantification for clinical risk communication.

1.6 Justification of the Study

1.6.1 Research Gap

The literature review confirms the existence of a specific, documented, and unresolved research gap. While maternal health prediction models exist globally, no study has built an individual-level, probabilistic maternal risk classification system specifically calibrated for Kenya's demographic profile using KDHS data. Existing Kenyan studies operate at county level (University of Nairobi, 2024), rely on hospital EHR data unavailable in community settings (Strathmore University preterm birth study), or use rule-based digital health tools without predictive analytics (mPAMANECH, Nairobi). Furthermore, the application of TabNet and TabTransformer to maternal health data, a recommendation emerging from the 2025 literature, has not yet been attempted on KDHS data in any published study.

1.6.2 Why the Recommended Approach

This study recommends a four-tier ML pipeline culminating in a stacking ensemble with Bayesian inference and SHAP explainability for the following evidence-based reasons:

- TabNet and TabTransformer are recommended over classical models because their attention mechanisms and contextual embeddings are architecturally suited to the predominantly categorical, high-dimensional structure of KDHS data, and have demonstrated superior AUC-ROC performance on comparable health survey datasets (Doshi et al., 2025; Sumon et al., 2025).

- A stacking ensemble is recommended because combining the outputs of multiple diverse base models consistently reduces variance and improves generalisability compared to any single model, a finding validated across multiple recent maternal health ML studies (Mamun et al., 2025; QEML-MHRC framework, 2024).

- Bayesian inference via PyMC is recommended because point-estimate predictions are insufficient for clinical decision support, a CHW needs to know not just that a woman is high risk, but how confident the model is in that classification. Posterior probability distributions with credible intervals provide this calibrated uncertainty, as validated by Sheridan et al. (2025) for pregnancy outcome prediction.

- AI API integration is recommended to bridge the gap between model output and CHW usability. Research by Okolo et al. (2024) confirmed that CHWs strongly prefer AI tools that explain their outputs, the AI API layer operationalises this finding by generating plain-language explanations in both English and Swahili, accessible to CHWs without data science training.

1.6.3 Significance of the Study

This study is significant at multiple levels. Clinically, it has the potential to reduce preventable maternal deaths by enabling timely referral of genuinely high-risk women before complications occur. At policy level, the county-level risk dashboard provides the Ministry of Health and NGOs with a real-time evidence base for resource allocation. Academically, the study makes an original methodological contribution by being the first to apply Bayesian-augmented TabNet/TabTransformer modelling to KDHS 2022 data

for CHW deployment. The study also contributes to the emerging body of literature on generative AI integration in community health worker decision support tools. It directly advances SDG 3.1 (reduce maternal mortality) and Kenya's Universal Health Coverage agenda.

1.7 Limitations of the Study

- **Cross-sectional data:** The KDHS 2022 is a cross-sectional survey, meaning it captures a single point-in-time snapshot of each woman's profile. Longitudinal changes in risk status during pregnancy cannot be tracked, and the model will not capture dynamic risk evolution.

- **Self-reported variables:** Several key variables, including number of ANC visits and pregnancy complications, are self-reported by respondents, introducing the possibility of recall bias and social desirability bias.

- **Target variable engineering:** The composite maternal risk target variable is constructed using a rule-based system derived from WHO clinical guidelines and literature, rather than observed clinical outcomes (e.g., mortality or severe morbidity). This represents a methodological limitation that future studies with linked health outcome data could address.

- **Generalisability:** The model is calibrated specifically for the Kenyan demographic and health profile as captured in the 2022 KDHS. Generalisability to other countries or future survey waves will require re-validation.

- **AI API dependency:** The AI API explanation layer requires internet connectivity and API credit availability, which may limit deployability in remote, low-connectivity settings, though this limitation does not affect the core ML model functionality.

- **No clinical validation:** This study does not include prospective clinical validation of the model's predictions against actual patient outcomes. Clinical validation would be required before any real-world deployment as a clinical decision aid.

1.8 Definition of Terms

Term	Definition
Maternal Mortality Ratio (MMR)	The number of maternal deaths per 100,000 live births in a given population, used as the primary indicator of maternal health system performance.
Community Health Worker (CHW)	A trained lay health worker who provides basic health services and health education to communities, particularly in rural and underserved areas. In Kenya, formally designated as Community Health Promoters (CHPs).
KDHS 2022	Kenya Demographic and Health Survey 2022, a nationally representative cross-sectional household survey conducted by KNBS and ICF International, covering health, demographic, and socioeconomic indicators for 17,414 Kenyan women aged 15–49.
Machine Learning (ML)	A branch of artificial intelligence in which computational models learn patterns from data to make predictions or decisions without being explicitly programmed for each outcome.
TabNet	A novel deep learning architecture for tabular data that uses sequential attention mechanisms to select relevant features at each decision step, providing built-in interpretability (Arik & Pfister, 2021).
TabTransformer	A deep learning architecture for tabular data that applies Transformer self-attention mechanisms to categorical feature embeddings, capturing contextual relationships between categorical variables (Huang et al., 2020).
Stacking Ensemble	A meta-learning technique that combines the predictions of multiple base models using a meta-learner (e.g., Logistic Regression) to produce a final, improved prediction.

Bayesian Inference	A statistical framework that updates prior beliefs about a parameter using observed data to produce a posterior probability distribution, enabling uncertainty quantification.
Credible Interval (CI)	The Bayesian equivalent of a confidence interval, a range within which the true parameter value falls with a specified probability (e.g., 95%) given the observed data and prior.
SHAP (SHapley Additive exPlanations)	A model-agnostic explainability method grounded in cooperative game theory that assigns each input feature a contribution value (SHAP value) for each individual prediction.
Antenatal Care (ANC)	Healthcare provided to pregnant women from conception to the onset of labour, including health assessments, risk screening, and health promotion.
eCHIS	Electronic Community Health Information System, Kenya's digital platform used by Community Health Promoters to record and report household health data.
AI API Integration	The use of a large language model API to generate natural language outputs (explanations, recommendations, letters) from structured model prediction outputs.
Optuna	A hyperparameter optimisation framework that uses Bayesian optimisation to efficiently search for the best model hyperparameters.

1.9 Chapter Summary

This chapter has introduced the research study on predicting maternal health risk among pregnant women in Kenya using machine learning and Bayesian inference. The background established the critical burden of maternal mortality in Kenya (355/100,000 live births) and the structural gap in CHW-facing data-driven risk tools. The problem statement identified the fourfold research gap: absence of individual-level ML prediction on

KDHS data, no novel tabular DL architecture applied to this domain, no Bayesian uncertainty quantification on Kenyan maternal data, and no AI API explanation layer for CHW guidance. The general objective, three specific objectives, and corresponding research questions were defined. The purpose, justification, and limitations were presented, followed by key term definitions. Chapter Two will provide a comprehensive review of the theoretical and empirical literature underpinning each research objective.

2 CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews the existing body of academic and grey literature relevant to the proposed study on predicting maternal health risk among pregnant women in Kenya using machine learning and Bayesian inference. The chapter is organised into four main sections: a general literature review contextualising the study within the broader landscape of maternal health, digital health innovation, and machine learning in healthcare; an empirical literature review structured around the three specific objectives of the study; a synthesis of research gaps that the proposed study fills; and a chapter summary. Literature reviewed spans peer-reviewed journal articles, conference papers, and industry reports published primarily between 2023 and 2026, with emphasis on 2025–2026 sources to ensure currency and relevance.

2.2 General Literature Review

2.2.1 Maternal Mortality — Global Burden and Kenyan Context

Maternal mortality remains one of the most persistent indicators of health system failure globally. The World Health Organization (2023) estimates the global maternal mortality ratio at 223 deaths per 100,000 live births, with Sub-Saharan Africa bearing a disproportionate burden, accounting for approximately 70% of global maternal deaths. In Kenya specifically, the KDHS 2022 places the maternal mortality ratio at 355 per 100,000 live births, significantly above the global average and nearly five times the SDG 3.1 target of 70 per 100,000 by 2030 (KNBS & ICF, 2023). The leading direct causes of maternal death in Kenya mirror global patterns: obstetric haemorrhage (27%), hypertensive disorders of pregnancy (14%), and sepsis (11%) account for the majority of deaths, with indirect causes, particularly anaemia (25% prevalence among pregnant women in the KDHS 2022) and HIV, compounding risk further in rural and low-wealth populations.

Ronsmans and Graham (2006), in their landmark Lancet review of maternal mortality epidemiology, established the foundational evidence base demonstrating that the majority of maternal deaths are preventable with access to skilled birth attendance and emergency obstetric care, findings that remain validated by more recent literature. Opoku

et al. (2025), in a scoping review protocol on risk prediction models for maternal and newborn outcomes in low- and middle-income countries published in PLOS ONE, concluded that the absence of validated, context-specific, individual-level risk prediction tools is a primary contributor to preventable maternal deaths across LMICs, directly motivating this study.

2.2.2 Kenya's Community Health System and Digital Health Landscape

Kenya's community health system has undergone significant transformation since 2020. The electronic Community Health Information System (eCHIS) has been deployed nationally, enabling over 100,000 Community Health Promoters to record household health data digitally. The Government of Kenya passed four landmark health laws in 2023, the Social Health Insurance Act, Digital Health Act, Primary Health Care Act, and Facility Improvement Financing Act, signalling deep policy commitment to digitally enabled UHC. Cabinet Secretary Aden Duale reaffirmed in early 2026 that digital innovation is a critical enabler of Kenya's UHC agenda (Ethical Business Africa, 2026).

Ombech et al. (2025), evaluating an integrated community-hospital digital health system for maternal and newborn care in northern Kenya, found statistically significant improvements in ANC uptake and skilled delivery rates following digital system introduction, but explicitly recommended the addition of ML-based predictive risk scoring as the critical next evolution of the eCHIS platform. This finding directly positions the proposed study as an implementation of their recommendation. PATH (2025), in a report on digital health in Kenya, similarly noted the absence of predictive analytics within eCHIS as a gap limiting proactive case management, validating the problem statement of this study.

2.2.3 Machine Learning in Healthcare — Evolution and Novel Architectures

The application of machine learning to healthcare prediction has grown exponentially. Vasudevan et al. (2025), in a scoping review of ML models for maternal morbidity and mortality prediction from electronic medical records published in the Journal of Medical Internet Research, identified 39 studies demonstrating ML feasibility across maternal outcomes. However, the review identified a critical gap: virtually all studies used hospital-based EHRs, making their models inapplicable to community settings. This gap directly

motivates the KDHS-based approach of the proposed study.

Classical ensemble methods, particularly Random Forest and XGBoost, have dominated health ML literature. However, 2025 has seen the emergence of architectures specifically designed for tabular (structured) health data that outperform classical methods. Doshi et al. (2025) demonstrated that TabNet and TabTransformer consistently outperformed Random Forest and XGBoost on AUC-ROC for toxicity prediction on tabular data. Sumon et al. (2025) proposed CardioTabNet, a hybrid TabTransformer for cardiac risk prediction, achieving superior sensitivity over tree-based models. Yasmeen et al. (2025) validated a hybrid TabNet stacking ensemble for heart disease prediction in Frontiers in Physiology, demonstrating that TabNet's sequential attention mechanism improves recall for minority classes, directly relevant to the expected class imbalance in the maternal risk target variable.

Critically, Tariq et al. (2026), published in BMC Pregnancy and Childbirth, specifically applied explainable tabular deep learning models, including TabNet, to antenatal caesarean delivery prediction in a multiparous population, reporting superior performance over classical models and recommending TabNet as the architecture of choice for structured obstetric prediction tasks. This study represents the closest existing methodological precursor to the proposed research, validating the architectural choice while leaving open the KDHS-specific, Bayesian-augmented, CHW-deployment application.

2.2.4 Explainable Artificial Intelligence in Clinical Settings

The adoption of machine learning in healthcare is contingent on clinical trust, which in turn requires model explainability. Abbas et al. (2025), in a meta-analysis of 62 studies on XAI in clinical decision support systems, confirmed SHAP as the most widely adopted and clinically useful explainability method for structured tabular data. Mamun et al. (2025) demonstrated the application of SHAP and LIME to a maternal health risk detection framework achieving 99.51% accuracy, establishing the current state of the art in explainable maternal health ML. Noor et al. (2025) identified the lack of CHW-oriented XAI design as a critical gap, motivating the proposed AI API explanation layer that translates SHAP outputs into plain language accessible to non-technical frontline workers.

Okolo et al. (2024), studying perceptions of XAI among CHWs in rural India, found

that while CHWs initially struggled with raw SHAP visualisations, they strongly preferred AI tools that included explanations over those that did not, underscoring the importance of a natural language interface between the model and the CHW. This finding directly supports the study's third objective.

2.3 Empirical Literature Review

2.3.1 Studies Related to Objective 1 — ML Classification for Maternal Risk

Khadidos et al. (2025), published in Healthcare (Vol. 13, No. 5), applied boosting-based machine learning models to high-risk pregnancy prediction using demographic and clinical variables comparable to those in KDHS. Their study found that gradient boosted models outperformed logistic regression on sensitivity, the priority metric for healthcare risk classification, and validated the clinical relevance of ANC visit frequency, maternal age, and parity as key predictors, confirming the feature set selected for the proposed study. However, their dataset was hospital-derived and excluded community-level wealth and socioeconomic indicators available in KDHS.

Tzimourta et al. (2025), in Procedia Computer Science, benchmarked multiple ML algorithms for maternal health risk factor prediction, reporting that ensemble models consistently outperformed single classifiers across precision, recall, and AUC metrics. Sarker et al. (2025), in Healthcare Analytics, proposed a PCA-enhanced XGBoost model with SMOTE-ENN for imbalanced maternal health data, achieving strong minority-class performance, directly relevant to the preprocessing pipeline of the proposed study. The Quad-Ensemble Machine Learning framework (QEML-MHRC, Scientific Reports, 2024) demonstrated that a four-model stacking architecture outperforms all component models individually, providing direct methodological precedent for the stacking ensemble proposed in this study.

Crucially, Tariq et al. (2026) applied TabNet specifically to obstetric prediction, reporting superior performance and recommending the architecture for structured maternal health data. No existing study has applied TabNet or TabTransformer to KDHS 2022 data at the individual level, this constitutes the primary methodological contribution of this study's first objective.

Studies recommend: Ensemble models for maternal risk prediction, SMOTE

for class imbalance, and TabNet/TabTransformer as superior architectures for tabular obstetric data. None has been applied to KDHS 2022 individual-level data.

2.3.2 Studies Related to Objective 2 — Bayesian Inference for Risk Quantification

Sheridan et al. (2025), published in Computers in Biology and Medicine, proposed a Bayesian Network model of pregnancy outcomes for England and Wales using publicly available population statistics, methodologically analogous to the KDHS data use proposed in this study. Their study demonstrated that Bayesian networks produce more reliable and clinically interpretable risk estimates than conventional ML classifiers for pregnancy outcomes, and validated the use of Bayesian models on national-level health survey data. The proposed study extends their approach by integrating Bayesian inference as an uncertainty quantification layer on top of an ML classification pipeline, a hybrid not yet applied to Kenyan maternal data.

Alhusseini et al. (2025) applied Bayesian Accelerated Failure Time modelling with Elastic Net regularisation to maternal mortality data, demonstrating that Bayesian methods with regularisation effectively address data sparsity in maternal health datasets. Their identification of fewer than four ANC visits and young maternal age as significant predictors directly validates the feature selection of the proposed study. Bodnar et al., in a related hierarchical Bayesian study, demonstrated that posterior probability outputs with credible intervals provide substantially more actionable clinical information than point predictions, validating the proposed study's use of PyMC to generate 95% credible intervals alongside each risk classification.

Studies recommend: Bayesian inference for calibrated uncertainty in maternal health prediction. No hybrid Bayesian-deep learning pipeline has been applied to KDHS 2022 data, this is the gap Objective 2 fills.

2.3.3 Studies Related to Objective 3 — AI-Powered CHW Decision Support

Ombech et al. (2025) explicitly recommended the addition of predictive analytics and AI-powered guidance to Kenya's eCHIS platform, the most direct institutional call for the intervention proposed in this study's third objective. MHP Salud (2025) documented

ASHABot, an LLM-powered chatbot developed for CHWs in India, as a pioneering example of generative AI deployment at the community health frontline, validating the technical feasibility of the proposed AI API integration. No equivalent tool exists for Kenyan CHWs in the maternal health domain.

PSO-optimised TabTransformer research (ScienceDirect, 2025) confirmed that 'the rapid advancement of large language models has significantly impacted healthcare, enabling automated clinical reasoning, medical text summarisation, and knowledge extraction', positioning AI API integration with ML prediction pipelines as the current research frontier. Noor et al. (2025) identified insufficient CHW-oriented AI design as a critical gap in health AI, specifically recommending bilingual (English and local language) interfaces for frontline health workers, the precise design choice implemented in this study's Swahili-language explanation feature.

> *Studies recommend: AI-powered explanation layers for CHW decision support and bilingual interfaces for LMIC settings. No such tool exists in Kenya for maternal health, this is the gap Objective 3 fills.*

2.4 Research Gaps

The literature review has identified the following specific, documented, and unresolved research gaps that this study addresses:

- **Gap 1:** No existing study has developed an individual-level maternal health risk classification model calibrated for Kenya using the KDHS 2022 dataset. Existing Kenyan ML studies operate at county level and cannot identify specific high-risk women.

- **Gap 2:** No existing study has applied novel tabular deep learning architectures, specifically TabNet and TabTransformer, to KDHS data for maternal risk prediction. This represents the primary methodological novelty of this study.

- **Gap 3:** No existing study has combined Bayesian inference (PyMC posterior probability distributions with credible intervals) with a deep learning tabular model pipeline for maternal health prediction on Kenyan national survey data.

- **Gap 4:** No deployed tool exists in Kenya that integrates an AI API explanation layer to generate plain-language maternal risk explanations, care recommendations, and referral letters for Community Health Workers in both English and Swahili.

2.5 Chapter Summary

This chapter has comprehensively reviewed the literature across four dimensions: the global and Kenyan maternal health burden, Kenya's digital health landscape, the evolution of machine learning and explainable AI in healthcare, and the specific empirical evidence base for each of the study's three objectives. The general literature review established that Kenya's 355/100,000 MMR is preventable with timely risk identification, that Kenya's eCHIS infrastructure is ready for a predictive ML layer, and that novel architectures (TabNet, TabTransformer) represent the current frontier of tabular health ML. The empirical review confirmed four specific research gaps, no KDHS individual-level ML, no TabNet on KDHS, no Bayesian-DL hybrid on Kenyan maternal data, and no AI API CHW explanation tool in Kenya, that this study fills. Chapter Three will detail the research methodology that operationalises these findings into a rigorous and implementable research design.

3 CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the research methodology for developing a machine learning-based maternal health risk prediction system using the Kenya Demographic and Health Survey 2022. The chapter begins with the design and experimental methodology, followed by the research design, a detailed treatment of the mathematical foundations and model equations underpinning the study, data collection methods and instrumentation, data analysis methods, validation and evaluation procedures, and ethical considerations governing the research.

3.2 Design and Experimental Methodology

This study adopts a quantitative, computational experimental methodology following the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework. The experimental design is comparative: multiple machine learning architectures are systematically trained and evaluated on identical train-validation-test splits to identify the model configuration that maximises predictive performance on the KDHS 2022 tabular dataset. The study is non-experimental with respect to human subjects, no primary data collection, participant recruitment, or variable manipulation is conducted. All experimentation occurs on anonymised secondary survey data.

The pipeline proceeds through five sequential phases: (1) data acquisition and preprocessing, (2) target variable engineering, (3) comparative model development (four tiers), (4) Bayesian inference integration, and (5) Streamlit application development with AI API integration. Each phase is designed to be reproducible, all code is version-controlled on GitHub and all model training is seeded with a fixed random state (42) to ensure deterministic reproducibility.

3.3 Research Design

The study adopts a non-experimental, quantitative secondary data analysis design. The KDHS 2022 is a cross-sectional survey, data reflects a single point-in-time snapshot of

each woman's demographic, socioeconomic, and reproductive health profile. The cross-sectional design is appropriate for the study's aim of building a community-level risk screening tool rather than a longitudinal clinical prognosis model. The ontological position is post-positivist, acknowledging that an objective maternal risk exists and can be approximated through data, while recognising model imperfection, which is why Bayesian uncertainty quantification is a core component.

3.4 Mathematical Foundations and Model Equations

This section presents the mathematical formulation of each model and analytical technique applied in this study. These equations constitute the theoretical and computational basis for all predictions, uncertainty estimates, and explainability outputs generated by the system.

3.4.1 Target Variable Engineering — Composite Risk Score

The target variable y is a three-class ordinal variable (0 = Low Risk, 1 = Moderate Risk, 2 = High Risk) engineered from eight binary clinical risk flags, each derived from KDHS variables using WHO evidence-based thresholds:

$$f_k(\mathbf{x}_i) = \begin{cases} 1 & \text{if condition}_k(\mathbf{x}_i) \text{ is satisfied} \\ 0 & \text{otherwise} \end{cases} \qquad \text{for } k \in \{1, \ldots, 8\} \tag{3.1}$$

where $k \in \{1, \ldots, 8\}$ denotes each of the eight risk conditions.

$$S_i = \sum_{k=1}^{8} f_k(\mathbf{x}_i) \tag{3.2}$$

S_i = composite risk score for woman i, ranging from 0 to 8.

$$y_i = \begin{cases} 0 \text{ (Low)} & \text{if } S_i = 0 \\ 1 \text{ (Moderate)} & \text{if } 1 \leq S_i \leq 2 \\ 2 \text{ (High)} & \text{if } S_i \geq 3 \end{cases} \tag{3.3a–c}$$

3.4.2 Data Preprocessing — SMOTE

Synthetic Minority Oversampling Technique (SMOTE) is applied to the training set to address class imbalance. For each minority class sample $\mathbf{x}_i$, SMOTE generates a synthetic sample $\mathbf{x}_{\text{new}}$ by interpolating between $\mathbf{x}_i$ and one of its k nearest neighbours $\mathbf{x}_{nn}$:

$$\mathbf{x}_{\text{new}} = \mathbf{x}_i + \lambda \cdot (\mathbf{x}_{nn} - \mathbf{x}_i), \quad \lambda \sim \text{Uniform}(0, 1) \tag{3.4}$$

where $\lambda \sim \text{Uniform}(0, 1)$ is a random interpolation coefficient, $k = 5$ nearest neighbours.

3.4.3 Tier 1 — Logistic Regression (Baseline)

For multinomial logistic regression with $C = 3$ classes, the probability of class c for observation $\mathbf{x}_i$ is modelled using the softmax function:

$$P(y_i = c \mid \mathbf{x}_i) = \frac{\exp(\mathbf{w}_c^T \mathbf{x}_i + b_c)}{\sum_{j=1}^{C} \exp(\mathbf{w}_j^T \mathbf{x}_i + b_j)} \tag{3.5}$$

where $\mathbf{w}_c \in \mathbb{R}^p$ are class-specific weight vectors, b_c are bias terms, p = number of features.

Parameters are estimated by minimising the categorical cross-entropy loss with L2 regularisation:

$$\mathcal{L}(\mathbf{W}, \mathbf{b}) = -\frac{1}{n} \sum_{i=1}^{n} \sum_{c=0}^{C-1} y_{ic} \cdot \log P(y_i = c \mid \mathbf{x}_i) + \frac{\lambda}{2} \|\mathbf{W}\|_F^2 \tag{3.6}$$

where n = sample size, y_{ic} is 1 if true class of observation i is c else 0, λ = regularisation strength.

3.4.4 Tier 2 — Random Forest

Random Forest is an ensemble of T decision trees, each trained on a bootstrap sample of the training data with a random subset of m features at each split. The final class prediction is the majority vote across all trees:

$$\hat{y}_i = \arg\max_c \sum_{t=1}^{T} \mathbb{I}(h_t(\mathbf{x}_i) = c) \tag{3.7}$$

where $h_t(\mathbf{x}_i)$ is the prediction of tree t for observation $\mathbf{x}_i$, $\mathbb{I}(\cdot)$ is the indicator function.

Each split node selects the feature j and threshold τ that maximises the reduction in Gini impurity:

$$\text{Gini}(D) = 1 - \sum_{c=1}^{C} p_c^2 \tag{3.8}$$

where p_c = proportion of class c samples in node dataset D.

$$\Delta\text{Gini}(j, \tau) = \text{Gini}(D) - \frac{|D_L|}{|D|} \cdot \text{Gini}(D_L) - \frac{|D_R|}{|D|} \cdot \text{Gini}(D_R) \tag{3.9}$$

where D_L, D_R are left/right child node datasets after splitting on (j, τ).

3.4.5 Tier 2 — XGBoost (Extreme Gradient Boosting)

XGBoost builds an additive ensemble of T regression trees, where the model at iteration t is:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta \cdot h_t(\mathbf{x}_i) \tag{3.10}$$

where $\eta \in (0, 1]$ is the learning rate, h_t is the t-th tree learned on the residuals.

At each iteration t, XGBoost minimises the regularised objective:

$$\text{Obj}^{(t)} = \sum_{i=1}^{n} \mathcal{L}(y_i, \hat{y}_i^{(t-1)} + h_t(\mathbf{x}_i)) + \Omega(h_t) \tag{3.11}$$

where $\Omega(h) = \gamma T + \frac{1}{2}\lambda\|\mathbf{w}\|^2$ penalises tree complexity (T = leaf count, $\mathbf{w}$ = leaf weights).

Using a second-order Taylor expansion of the loss function:

$$\text{Obj}^{(t)} \approx \sum_{i=1}^{n} \left[g_i \cdot h_t(\mathbf{x}_i) + \frac{1}{2} h_i \cdot h_t(\mathbf{x}_i)^2 \right] + \Omega(h_t) \tag{3.12}$$

where $g_i = \partial\mathcal{L}/\partial\hat{y}^{(t-1)}$ (first-order gradient), $h_i = \partial^2\mathcal{L}/\partial(\hat{y}^{(t-1)})^2$ (second-order Hessian).

3.4.6 Tier 3 — TabNet (Novel Architecture)

TabNet uses sequential multi-step attention to select relevant features at each decision step. Let $\mathbf{a}^{[i]} \in \mathbb{R}^p$ be the processed features for observation i. At step s, the attention mask $\mathbf{M}^{[s]}$ is computed as:

$$\mathbf{M}^{[s]}(i) = \text{sparsemax}\left(\mathbf{P}^{[s]}(i) \odot \mathbf{h}_{\text{att}}(\mathbf{a}^{[s]}(i))\right) \tag{3.13}$$

where sparsemax projects the attention logits to a sparse probability simplex, $\mathbf{P}^{[s]}$ is the prior scale.

The prior scale $\mathbf{P}^{[s]}$ penalises features already used in previous steps:

$$\mathbf{P}^{[s]}(i) = \prod_{s'=1}^{s-1} \left(\gamma_{\text{att}} - \mathbf{M}^{[s']}(i)\right) \tag{3.14}$$

where $\gamma_{\text{att}} \in \mathbb{R}$ is a coefficient controlling feature re-usage (hyperparameter).

The masked features are fed into a feature transformer block (shared + step-specific FC layers with batch normalisation and GLU activations):

$$\mathbf{h}^{[s]}(i) = f_{\text{step}}\left(\mathbf{M}^{[s]}(i) \odot \mathbf{a}^{[i]}\right) \tag{3.15}$$

where $\odot$ denotes element-wise multiplication, f_{step} is the step-specific transformer block.

The final output aggregates representations across all N_{steps} decision steps:

$$\text{Output}(i) = \sum_{s=1}^{N_{\text{steps}}} \text{ReLU}\left(\mathbf{h}^{[s]}(i)\right) \tag{3.16}$$

This is passed to the final classification head with softmax for class probability prediction.

The global feature importance of feature j from TabNet's attention masks is:

$$\text{Importance}_j = \sum_{s=1}^{N_{\text{steps}}} \sum_{i=1}^{n} \eta_s \cdot \mathbf{M}_j^{[s]}(i) \tag{3.17}$$

where η_s is the step-level coefficient from the step's contribution to the output.

3.4.7 Tier 3 — TabTransformer (Novel Architecture)

TabTransformer embeds each categorical feature c_j into a d-dimensional space using a learnable embedding matrix $\mathbf{E}_j \in \mathbb{R}^{|C_j| \times d}$:

$$\mathbf{e}_j = \mathbf{E}_j[c_j] \quad \forall j \in \{1, \ldots, m_{\text{cat}}\} \tag{3.18}$$

where m_{cat} = number of categorical features, $|C_j|$ = cardinality of feature j.

These embeddings are passed through L Transformer encoder layers. Each layer applies multi-head self-attention (MHSA) with H heads:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{3.19}$$

where $\mathbf{Q} = \mathbf{X}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{X}\mathbf{W}_K$, $\mathbf{V} = \mathbf{X}\mathbf{W}_V$ are query/key/value projections, $d_k = d/H$.

$$\text{MHSA}(\mathbf{X}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_H) \cdot \mathbf{W}_O \tag{3.20}$$

where $\text{head}_h = \text{Attention}(\mathbf{X}\mathbf{W}_Q^h, \mathbf{X}\mathbf{W}_K^h, \mathbf{X}\mathbf{W}_V^h)$.

Each encoder layer applies layer normalisation and a feed-forward network:

$$\mathbf{z} = \text{LayerNorm}(\mathbf{X} + \text{MHSA}(\mathbf{X})) \tag{3.21}$$

$$\text{Output} = \text{LayerNorm}(\mathbf{z} + \text{FFN}(\mathbf{z})) \tag{3.22}$$

$\text{FFN}(\mathbf{z}) = \max(0, \mathbf{z}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2$ (two-layer MLP with ReLU).

Contextual categorical embeddings from the final Transformer layer are concatenated with continuous features and passed to an MLP classifier:

$$\mathbf{x}_{\text{MLP}} = [\mathbf{e}_1^L \parallel \mathbf{e}_2^L \parallel \cdots \parallel \mathbf{e}_{m_{\text{cat}}}^L \parallel \mathbf{x}_{\text{cont}}] \tag{3.23}$$

where $\parallel$ denotes concatenation, $\mathbf{e}_j^L$ = final layer embedding of categorical feature j, $\mathbf{x}_{\text{cont}}$ = continuous features.

3.4.8 Tier 4 — Stacking Ensemble

The stacking ensemble uses predictions from TabNet, XGBoost, and CatBoost as input features to a Logistic Regression meta-learner. For observation $\mathbf{x}_i$:

$$\mathbf{z}_i = [\mathbf{P}_{\text{TabNet}}(y \mid \mathbf{x}_i) \parallel \mathbf{P}_{\text{XGB}}(y \mid \mathbf{x}_i) \parallel \mathbf{P}_{\text{CatBoost}}(y \mid \mathbf{x}_i)] \tag{3.24}$$

where $\mathbf{P}_M(y \mid \mathbf{x}_i) \in \mathbb{R}^3$ is the class probability vector from model M for observation i.

$$\hat{y}_i^{\text{stack}} = \arg\max_c \left[\text{softmax}\left(\mathbf{W}_{\text{meta}} \cdot \mathbf{z}_i + \mathbf{b}_{\text{meta}}\right)\right]_c \tag{3.25}$$

where $\mathbf{W}_{\text{meta}}, \mathbf{b}_{\text{meta}}$ are the meta-learner parameters estimated by cross-validated logistic regression.

To prevent data leakage, meta-features $\mathbf{z}_i$ are generated using out-of-fold predictions from 5-fold stratified cross-validation on the training set, then the full training set is used to retrain the base models before predicting on the test set.

3.4.9 Bayesian Inference — Posterior Distribution

Following model selection, Bayesian logistic regression is applied using PyMC to quantify predictive uncertainty. For each class c, the model places a weakly informative Normal prior on the coefficient vector $\boldsymbol{\beta}_c$ and intercept α_c:

$$\alpha_c \sim \mathcal{N}(0, 1) \tag{3.26}$$

$$\boldsymbol{\beta}_c \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad \text{for } c \in \{0, 1, 2\} \tag{3.27}$$

Weakly informative priors centre predictions near 0 while allowing data to dominate. The likelihood is modelled as multinomial logistic regression:

$$P(y_i = c \mid \mathbf{x}_i, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \frac{\exp(\alpha_c + \boldsymbol{\beta}_c^T \mathbf{x}_i)}{\sum_{j=0}^{2} \exp(\alpha_j + \boldsymbol{\beta}_j^T \mathbf{x}_i)} \tag{3.28}$$

Applying Bayes' theorem:

$$\pi(\boldsymbol{\alpha}, \boldsymbol{\beta} \mid \text{data}) \propto \mathcal{L}(\text{data} \mid \boldsymbol{\alpha}, \boldsymbol{\beta}) \cdot \pi(\boldsymbol{\alpha}) \cdot \pi(\boldsymbol{\beta}) \tag{1}$$

The posterior distribution is approximated using Markov Chain Monte Carlo (MCMC) with the No-U-Turn Sampler (NUTS), the default sampler in PyMC v5.x, with 2,000 tuning steps and 2,000 draw steps. The posterior predictive distribution for a new observation $\mathbf{x}_*$ is:

$$P(y_* = c \mid \mathbf{x}_*, \text{data}) = \int P(y_* = c \mid \mathbf{x}_*, \boldsymbol{\alpha}, \boldsymbol{\beta}) \cdot \pi(\boldsymbol{\alpha}, \boldsymbol{\beta} \mid \text{data})\, d\boldsymbol{\alpha}\, d\boldsymbol{\beta} \tag{3.29}$$

Approximated by Monte Carlo integration over posterior samples:

$$\frac{1}{S} \sum_{s=1}^{S} P(y_* = c \mid \mathbf{x}_*, \boldsymbol{\alpha}^s, \boldsymbol{\beta}^s) \tag{2}$$

The 95% Highest Density Interval (HDI) for the High Risk class probability provides the credible interval displayed to CHWs:

$$\text{HDI}_{0.95} = \left[q_{0.025} \left(P(y_* = 2 \mid \mathbf{x}_*, \text{data}) \right),\ q_{0.975} \left(P(y_* = 2 \mid \mathbf{x}_*, \text{data}) \right) \right] \tag{3.30}$$

where q_α denotes the α-quantile of the posterior predictive distribution.

3.4.10 SHAP — Feature Attribution

SHAP values are grounded in Shapley values from cooperative game theory. For observation $\mathbf{x}_i$ with feature set F, the SHAP value ϕ_j for feature j is:

$$\phi_j(f, \mathbf{x}_i) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \cdot \left[f(S \cup \{j\}) - f(S) \right] \tag{3.31}$$

where S is a subset of features excluding j, $f(S)$ is the model prediction using only features in S.

For tree-based models (Random Forest, XGBoost, CatBoost), TreeSHAP computes exact SHAP values in $O(TLD^2)$ time, where T = number of trees, L = maximum leaves, D = maximum depth. The global feature importance for feature j is the mean absolute SHAP value across all observations:

$$I_j = \frac{1}{n} \sum_{i=1}^{n} |\phi_j(f, \mathbf{x}_i)| \tag{3.32}$$

Global importance ranks features by their average contribution magnitude across all predictions.

3.4.11 Evaluation Metrics

For $C = 3$ classes, with TP_c, FP_c, FN_c denoting true positives, false positives, and false negatives for class c:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \tag{3.33}$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{3.34}$$

$$F1_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{3.35}$$

$$\text{Macro-Precision} = \frac{1}{C} \sum_{c=1}^{C} \text{Precision}_c \tag{3.36}$$

$$\text{Macro-Recall} = \frac{1}{C} \sum_{c=1}^{C} \text{Recall}_c \tag{3.37}$$

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^{C} F1_c \tag{3.38}$$

The Matthews Correlation Coefficient (MCC) for multi-class is computed as:

$$\text{MCC} = \frac{\sum_k \sum_l \sum_m C_{kk} C_{lm} - C_{kl} C_{mk}}{\sqrt{\left(\sum_k \left(\sum_l C_{kl}\right) \left(\sum_l C_{lk}\right)\right) \cdot \left(\sum_k \left(\sum_l C_{lk}\right) \left(\sum_l C_{kl}\right)\right)}} \tag{3.39}$$

where $\mathbf{C}$ is the $C \times C$ confusion matrix; MCC $\in [-1, 1]$, with 1 indicating perfect prediction.

The Brier Score for probabilistic calibration assessment:

$$\text{BS} = \frac{1}{n} \sum_{i=1}^{n} \sum_{c=0}^{C-1} \left(P(y_i = c \mid \mathbf{x}_i) - \mathbb{I}(y_i = c)\right)^2 \tag{3.40}$$

Lower Brier Score indicates better-calibrated probability outputs; BS = 0 is perfect calibration.

3.5 Data Collection Methods

3.5.1 Primary Data Source

This study uses the Kenya Demographic and Health Survey 2022 (KDHS 2022) as its sole data source, the Individual Recode (IR) file accessed through the DHS Program portal (dhsprogram.com) following institutional registration and data use agreement approval. The KDHS 2022 used a stratified two-stage cluster sampling design covering all 47 counties of Kenya, yielding 17,414 completed women's questionnaires. The IR file contains 32,156 records (reflecting multiple birth-history rows per woman) and 5,925 variables.

3.5.2 Variable Selection and Dataset Characteristics

Thirteen variables were selected based on clinical evidence, WHO antenatal care guidelines, and data availability:

Table 2: Variables Selected for the Study

Code	Variable	Type	Clinical Justification
v012	**Maternal age**	Continuous	Adolescent (¡18) and advanced (¿35) age elevate obstetric risk
v201	**Parity (total births)**	Continuous	High parity ($\geq$ 5) associated with haemorrhage and uterine atony
v106	**Education level**	Ordinal (0–3)	Lower education reduces health-seeking behaviour
v190	**Wealth index**	Ordinal (1–5)	Poverty is primary barrier to facility delivery
v025	**Urban/rural**	Binary	Rural residence limits emergency obstetric access
v501	**Marital status**	Categorical	Social support influences ANC attendance

v212	**Age at first birth**	Continuous	Early first birth compounds cumulative risk
v024	**County/region**	Categorical	County-level disparities in outcomes well documented
m14_1	**ANC visits (recent birth)**	Continuous	¡4 visits is WHO-defined risk indicator
m15_1	**Place of delivery**	Categorical	Home delivery elevates complication risk
m17_1	**Caesarean section**	Binary	Prior CS affects subsequent delivery profile
m3a_1	**Pregnancy complication**	Binary	Self-reported complication is direct risk indicator
b11_01	**Birth interval (months)**	Continuous	¡24 months associated with adverse maternal outcomes

3.6 Instrumentation

All analysis is conducted in Python 3.11 using Google Colab as the cloud computing environment. The following libraries constitute the instrumentation of this study:

Table 3: Software Libraries and Tools Used

Library/Tool	Purpose	Version
pandas	Data loading, manipulation, and feature engineering	2.x
numpy	Numerical computation and array operations	1.26.x
scikit-learn	Preprocessing, classical ML models, cross-validation, metrics	1.4.x
imbalanced-learn	SMOTE implementation for class imbalance handling	0.12.x
pytorch-tabnet	TabNet architecture implementation	4.1.x

tab_transformer_pytorch	TabTransformer architecture implementation	0.3.x
xgboost	XGBoost classifier implementation	2.x
catboost	CatBoost classifier (native categorical support)	1.2.x
pymc	Bayesian inference and MCMC posterior sampling	5.x
arviz	Bayesian posterior diagnostics and visualisation	0.18.x
shap	SHAP value computation for model explainability	0.45.x
optuna	Bayesian hyperparameter optimisation (100 trials)	3.x
streamlit	Web application framework for end product	1.3x
matplotlib / seaborn	Data visualisation and EDA	3.8.x / 0.13.x
Google Colab	Cloud GPU-enabled Python notebook environment	—
GitHub	Code version control and reproducibility	—

3.7 Method of Data Analysis

3.7.1 Exploratory Data Analysis

Prior to modelling, comprehensive EDA will be conducted including: univariate frequency distributions and descriptive statistics; bivariate analysis using chi-square tests (categorical-target associations) and ANOVA (continuous-target associations); correlation heatmap for multicollinearity detection; missing value profiling; and class distribution analysis of the engineered target variable.

3.7.2 Preprocessing Pipeline

The preprocessing pipeline consists of: (1) median imputation for v212 (33 missing values); (2) zero imputation for b11_01 (2,835 missing, first births); (3) Label Encoding for ordinal and binary variables; (4) StandardScaler for continuous variables; (5) 70/15/15 stratified train-validation-test split; and (6) SMOTE applied exclusively to the training set.

3.7.3 Model Training and Hyperparameter Optimisation

All Tier 2–4 models undergo hyperparameter optimisation using Optuna with 100 trials and 5-fold stratified cross-validation. The objective function maximises macro AUC-ROC on the validation fold. The final test set is used only once after model selection is complete to provide an unbiased performance estimate.

3.7.4 Feature Importance Analysis

Feature importance is assessed using: (1) tree model built-in importance (Gini impurity reduction); (2) SHAP global summary plots (mean |SHAP value| per feature); (3) SHAP waterfall plots for individual predictions; and (4) TabNet's built-in attention mask feature importance.

3.8 Validity and Evaluation

Internal validity is ensured through 5-fold stratified cross-validation for all model comparisons, fixed random seeds (42) for reproducibility, and hold-out test set evaluation. External validity is addressed through the nationally representative KDHS 2022 sampling design, which covers all 47 Kenyan counties with probability proportional to size cluster sampling.

Model evaluation uses six metrics appropriate for multi-class imbalanced classification: macro-averaged Precision, Recall, F1-Score, MCC, macro AUC-ROC (one-vs-rest), and Brier Score for calibration. Priority is placed on High Risk class Recall, a missed high-risk woman is the costliest classification error. Application validation includes technical validation (50 synthetic patient profiles) and content validation (supervisor review of AI API outputs).

3.9 Ethical Considerations

- NACOSTI and IRB approval will be obtained from the National Commission for Science, Technology and Innovation (NACOSTI) and the USIU-Africa Institutional Review Board (IRB) prior to commencement of the implementation phase.
- The KDHS 2022 dataset is fully de-identified by KNBS and ICF International. No personally identifiable information is present. The DHS Program data use agreement strictly prohibits redistribution or re-identification of records.
- The AI API receives only computed risk scores and SHAP values, no individual KDHS records are transmitted to external APIs.
- All model results will be reported transparently including failure cases and below-expectation performance. All code will be publicly available on GitHub upon project completion.
- The application includes a disclaimer clearly stating that predictions are clinical decision support tools and must be interpreted alongside CHW professional judgement, not as a substitute for clinical assessment.

3.10 Chapter Summary

This chapter has presented the full research methodology for the proposed study. The non-experimental, CRISP-DM-structured design is grounded in the KDHS 2022 cross-sectional survey dataset. The mathematical foundations section detailed the equations governing all seven model architectures, Bayesian inference, SHAP explainability, SMOTE preprocessing, and six evaluation metrics. The four-tier model pipeline, from logistic regression baseline through novel TabNet/TabTransformer architectures to a Bayesian-augmented stacking ensemble, constitutes both a methodological contribution and a practical deployment system. Ethical compliance under NACOSTI and USIU-Africa IRB oversight ensures the research meets institutional and national standards. Chapter Four (in the full implementation report) will present the results of model training and evaluation.

References

Abbas, Q., Jeong, W., & Lee, S. W. (2025). Explainable AI in clinical decision support systems: A meta-analysis of methods, applications, and usability challenges. *Healthcare, 13*(17), 2154. https://doi.org/10.3390/healthcare13172154

Alhusseini, N., et al. (2025). Bayesian accelerated failure time models with elastic net regularisation for maternal mortality risk. *Taylor & Francis Online*. https://doi.org/10.1080/02664763.2

Arik, S. Ö., & Pfister, T. (2021). TabNet: Attentive interpretable tabular learning. *Proceedings of the AAAI Conference on Artificial Intelligence, 35*(8), 6679–6687.

Bodnar, L. M., et al. (2022). Bayesian hierarchical logistic regression for preterm birth risk: A methodological framework. *American Journal of Epidemiology, 191*(2), 340–352.

Doshi, H., et al. (2025). TabNet and TabTransformer: Novel deep learning models for chemical toxicity prediction in comparison with machine learning. *Journal of Applied Toxicology, 45*(9), 1730–1749. https://doi.org/10.1002/jat.4803

Ethical Business Africa. (2026, January 8). Innovation is improving maternal health in Kenya. https://ethicalbusiness.africa

Huang, X., Khetan, A., Cvitkovic, M., & Karnin, Z. (2020). TabTransformer: Tabular data modeling using contextual embeddings. *arXiv preprint arXiv:2012.06678*.

Kenya National Bureau of Statistics (KNBS) & ICF. (2023). *Kenya Demographic and Health Survey 2022*. KNBS and ICF, Nairobi, Kenya.

Khadidos, A. O., et al. (2025). Prediction of high-risk pregnancy using machine learning algorithms. *Healthcare, 13*(5). https://doi.org/10.3390/healthcare13050xxx

Mamun, M., et al. (2025). Identification of maternal health risk from optimal features using explainable machine learning. *Engineering Reports*, Wiley. https://doi.org/10.1002/eng2.70491

MHP Salud. (2025, December). Utilizing AI tools to amplify community health workers' impact. https://mhpsalud.org

Noor, et al. (2025). Unveiling explainable AI in healthcare: Current trends, challenges, and future directions. *WIREs Data Mining and Knowledge Discovery*. https://doi.org/10.1002/v

Okolo, C. T., Agarwal, D., Dell, N., & Vashistha, A. (2024). Examining perceptions of explainable AI with community health workers in rural India. *Proceedings of the ACM on Human-Computer Interaction, 8*(CSCW1), 71.

Ombech, E. A., et al. (2025). Effectiveness of an integrated community and hospi-

tal digital health information system for maternal and newborn healthcare in northern Kenya. *Frontiers in Digital Health, 7*, 1584733. https://doi.org/10.3389/fdgth.2025.1584733

Opoku, D. A., et al. (2025). Risk prediction models to determine maternal and newborn adverse pregnancy outcomes in low and middle-income countries: A scoping review protocol. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0318658

PATH. (2025). Strengthening community health through digital innovation in Kenya. https://www.path.org

Ronsmans, C., & Graham, W. J. (2006). Maternal mortality: Who, when, where, and why. *The Lancet, 368*(9542), 1189–1200. https://doi.org/10.1016/S0140-6736(06)69380-X

Sarker, M., et al. (2025). Predicting maternal health risk using PCA-enhanced XGBoost and SMOTE-ENN. *Healthcare Analytics, 5*, 100368.

Say, L., Chou, D., Gemmill, A., et al. (2014). Global causes of maternal death: A WHO systematic analysis. *The Lancet Global Health, 2*(6), e323–e333. https://doi.org/10.1016/S2214-109X(14)70227-X

Sheridan, M., et al. (2025). A Bayesian Network model of pregnancy outcomes for England and Wales. *Computers in Biology and Medicine, 189*, 109975. https://doi.org/10.1016/j.compb

Sumon, M. S. I., et al. (2025). CardioTabNet: A novel hybrid transformer model for heart disease prediction using tabular medical data. *Health Information Science and Systems, 13*, 44. https://doi.org/10.1007/s13755-025-00361-7

Tariq, A., et al. (2026). Explainable tabular deep learning models for antenatal cesarean delivery prediction in multiparous women. *BMC Pregnancy and Childbirth*. https://doi.org/10.1186/s12884-026-08934-4

Tzimourta, K. D., et al. (2025). Prediction of maternal health risk factors using machine learning algorithms. *Procedia Computer Science, 240*, 1234–1242.

Vasudevan, L., et al. (2025). Machine learning models to predict risk of maternal morbidity and mortality from electronic medical record data: Scoping review. *Journal of Medical Internet Research, 27*, e68225. https://doi.org/10.2196/68225

World Health Organization. (2023). *Trends in maternal mortality estimates 2000 to 2023*. WHO, Geneva.

Yasmeen, R., Khan, L., & Choi, A. (2025). Heart disease prediction using hybrid TabNet architecture with stacked ensemble learning. *Frontiers in Physiology, 16*, 1665128.

https://doi.org/10.3389/fphys.2025.1665128

APPENDICES

Appendix A: Introduction Letter

Date: 17/08/2026

To: The Head of Department,

From: Selmah Tzindori Mise
Student, B.Sc. Data Science and Analytics
United States International University – Africa (USIU-Africa)
ID: 669602

Subject: Introduction Letter — Research Project on Maternal Health Risk Prediction

Dear Sir/Madam,

I am writing to formally introduce myself as a Senior student in the Bachelor of Science in Data Science and Analytics programme at the United States International University – Africa (USIU-Africa), Nairobi, Kenya. I am currently conducting my final year research project as part of the requirements for the completion of my degree.

My research project is titled:

"Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference on the Kenya Demographic and Health Survey 2022"

The primary objective of this research is to develop a machine learning-based maternal health risk prediction system using the KDHS 2022 Individual Recode dataset. The system will classify pregnant women as Low, Moderate, or High risk based on demographic, socioeconomic, and reproductive health variables. The research will employ a four-tier machine learning pipeline incorporating classical ensemble methods, novel tabular deep learning architectures (TabNet and TabTransformer), Bayesian inference for uncertainty quantification, and SHAP for model explainability. The end product will be a Streamlit web application with AI API integration for Community Health Workers in Kenya.

This research is supervised by **Dr. Gabriel Okello** (gokello@usiu.ac.ke) of the Department of Data Science and Analytics at USIU-Africa. The research proposal is currently being submitted for review and approval in accordance with the relevant institutional and regulatory guidelines.

I kindly request your consideration and approval of this research project. I am available to provide any additional information as required.

Yours sincerely,

Student Signature: *S.T.M* Date: 17/08/2026

Supervisor Signature: _____ Date: _____

Department Head Signature: _____ Date: _____

Appendix B: Budget

Table 4: Research Budget

Item	**Purpose**	**Cost (USD)**	**Cost (KES)**
AI API Credits (Anthropic)	AI explanation layer, referral letters, risk explanations	$5.00	KES 645
Google Colab (free tier)	Cloud ML training environment	$0.00	KES 0
Streamlit Community Cloud	Web application deployment	$0.00	KES 0
KDHS 2022 Data Access (DHS Program)	Primary dataset (academic, free)	$0.00	KES 0
GitHub (free tier)	Code version control and open science	$0.00	KES 0
NACOSTI Application Fee	National ethical clearance application	—	KES 1,000
USIU-Africa IRB Application	Institutional ethical clearance	—	KES 3,000
Printing and Binding	Final proposal and report hard copies	—	KES 1,500
Internet Data (12 months)	Literature search, Colab, API access	—	KES 2,000
Miscellaneous	Stationery, USB storage backup	—	KES 500
TOTAL		**$\tilde{\$}$5.00**	**KES 8,645**

Appendix C: Research Workplan and Timeline

The following Gantt chart outlines the proposed timeline for the research project, spanning the Summer Semester 2026 (Proposal Phase) and Fall Semester 2026 (Implementation Phase).

Table 5: Research Workplan — Summer & Fall Semester 2026

Activity	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**
PHASE 1 — PROPOSAL (Summer Semester 2026, Weeks 1–7)							
Topic + Problem Statement + Literature Review	•	•					
Dataset Access + Methodology Draft			•	•			
Expected Results + Scope + Gantt					•		
Concept Note Presentation						•	
Full Proposal + NACOSTI/IRB Submission							•
Proposal Defense & Submission							•
PHASE 2 — IMPLEMENTATION (Fall Semester 2026, Weeks 1–7)							
Activity	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**
Data Preprocessing + EDA + Feature Engineering	•	•					
Baseline + Advanced ML Models + Optuna Tuning			•	•			

Bayesian Inference + SHAP + AI API Integration					•	•	
Streamlit App + Testing + Final Report + Defense							•

Key Milestones:

- **Summer Semester 2026 (Weeks 1–7):**
  - Week 1–2: Topic development, problem statement, literature review
  - Week 3–4: Dataset access and methodology drafting
  - Week 5: Expected results, scope definition, and Gantt chart
  - Week 6: Concept note presentation
  - Week 7: Full proposal submission, NACOSTI/IRB application, and proposal defense

- **Fall Semester 2026 (Weeks 1–7):**
  - Week 1–2: Data preprocessing, exploratory data analysis (EDA), and feature engineering
  - Week 3–4: Baseline model development, advanced ML models (TabNet, TabTransformer, XGBoost, CatBoost), and Optuna hyperparameter tuning
  - Week 5–6: Bayesian inference with PyMC, SHAP explainability implementation, and AI API integration
  - Week 7: Streamlit web application development, testing, final report writing, and project defense

Total Duration: 14 Weeks (Summer Semester 2026: Weeks 1–7 + Fall Semester 2026: Weeks 1–7)

Appendix D: Data Collection Sheet / Variable Codebook

The following Python code constitutes the data extraction and collection instrument for this study, applied to the KDHS 2022 Individual Recode (IR) file (KEIR8CFL.DTA):

```
import pandas as pd

df = pd.read_stata('KEIR8CFL.DTA', convert_categoricals=False)

columns_needed = ['v012','v201','v106','v190','v025','v501','v212',
                  'v024','m14_1','m15_1','m17_1','m3a_1','b11_01']

maternal_df = df[columns_needed].copy()

maternal_df = maternal_df.dropna(subset=['m14_1','m15_1','m17_1'])

# Resulting dataset: 10,391 records × 13 variables after filtering to
# women with recent birth records.
```

Appendix E: Experimental Protocols

Protocol 1 — Model Training

- All models trained in Google Colab with Python 3.11, random_state=42 for reproducibility.
- Dataset split: 70% training (7,274 records), 15% validation (1,559), 15% test (1,558) using stratified split.
- SMOTE applied only to training set using $k = 5$ nearest neighbours to balance class distribution.
- Hyperparameter optimisation: Optuna TPE sampler, 100 trials, 5-fold stratified cross-validation, maximising macro AUC-ROC.
- Test set used only once after final model selection to prevent data leakage.

Protocol 2 — Bayesian Inference

- PyMC v5.x with NUTS sampler: 4 chains, 2,000 tuning steps, 2,000 draw steps per chain.
- Convergence assessed using $\hat{R}$ statistic (target: $\hat{R} < 1.01$) and effective sample size (ESS > 400 per parameter).
- Posterior predictive checks performed to validate Bayesian model fit.

Protocol 3 — Application Validation

- Technical validation: 50 synthetic patient profiles spanning all three risk classes tested through the Streamlit application.
- Content validation: AI API explanation outputs reviewed by the supervisor for clinical coherence and language accuracy.
- Bilingual validation: Swahili outputs reviewed for grammatical accuracy and clinical relevance.

Appendix F: Dataset Description

- **Dataset:** Kenya Demographic and Health Survey 2022 — Individual Recode (IR)
- **Source:** Kenya National Bureau of Statistics (KNBS) and ICF International
- **Access:** DHS Program Portal — dhsprogram.com (requires registration and data use agreement approval)
- **File:** KEIR8CFL.DTA (Stata format) | Size: ∼180 MB | Original dimensions: 32,156 rows × 5,925 columns
- **Working dataset after filtering:** 10,391 rows × 13 columns
- **Target variable:** Composite maternal risk score (engineered — Low/Moderate/High)
- **Survey period:** 2022–2023 | Geographic coverage: All 47 counties of Kenya
- **Sampling design:** Stratified two-stage cluster sampling, probability proportional to size
- **Ethical status:** Fully de-identified — no PII present. Public access under DHS Program data use agreement.

Appendix G: Model and Algorithm Specifications

Table 6: Model Hyperparameter Specifications

Model	**Key Hyperparameters**	**Optuna Search Range**	**Library**
Logistic Regression	C (regularisation)	[0.01 – 100]	sklearn
Decision Tree	max_depth, min_samples_split	depth: [3–20], split: [2–20]	sklearn
Random Forest	n_estimators, max_depth, max_features	trees: [100–500], depth: [5–30]	sklearn
XGBoost	n_estimators, learning_rate, max_depth, subsample	lr: [0.01–0.3], depth: [3–12]	xgboost
CatBoost	iterations, learning_rate, depth	iterations: [100–1000], lr: [0.01–0.3]	catboost
TabNet	N_a, N_steps, gamma, lambda_sparse	N_a: [8–64], N_steps: [3–10]	pytorch-tabnet
TabTransformer	n_layers, n_heads, dim_head, mlp_dim	layers: [1–6], heads: [4–16]	tab_transformer_pytorch
Stacking Ensemble	Meta-learner: Logistic Regression C	C: [0.01–10]	sklearn

Appendix H: Informed Consent / Assent Form

NOTE: This study uses fully anonymised secondary data (KDHS 2022) and does not involve direct human participant recruitment. No individual consent is required for the core analysis. This consent form applies to any future pilot testing of the Streamlit application with CHW participants.

PARTICIPANT INFORMED CONSENT FORM

Study Title: Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference

Principal Investigator: Selmah Tzindori Mise | USIU-Africa | ID: 669602

Contact: [stmise@usiu.ac.ke] | Supervisor: Dr. Gabriel Okello (`gokello@usiu.ac.ke`)

Purpose of the Study:

You are being invited to participate in the user testing phase of a maternal health risk prediction web application. The application is designed to support Community Health Workers in assessing maternal health risk. Your participation involves interacting with the application interface and providing feedback on its usability and clarity.

What Participation Involves:

- You will be asked to enter sample patient data into the application and review the generated risk scores, explanations, and referral letters.
- The session will last approximately 30–45 minutes.
- No real patient data will be used. All test data is synthetic.
- No personally identifiable information about you will be collected or stored.

Risks and Benefits:

There are no anticipated risks associated with participation. Your feedback will contribute to improving a tool designed to reduce preventable maternal deaths in Kenya.

Voluntary Participation:

Participation is entirely voluntary. You may withdraw at any time without consequence.

Confidentiality:

All feedback data will be stored securely and reported in aggregate form only. No individual responses will be identifiable in any publication or report.

I have read and understood the above information. I voluntarily agree to participate in this study.

Participant Signature: _____ Date: _____

Participant Name (Print): _____ Date: _____

Researcher Signature: *S.T.M* Date: 17/08/2026

Appendix I: Debrief Form

PARTICIPANT DEBRIEF FORM

Study Title: Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference

Principal Investigator: Selmah Tzindori Mise | USIU-Africa | ID: 669602

Thank you for participating in the user testing session for the Maternal Risk Prediction Application. This form provides a full explanation of the study you have participated in.

Full Purpose of the Study:

This study aims to develop a machine learning-based tool that predicts maternal health risk for individual pregnant women in Kenya, to support Community Health Workers in identifying and referring high-risk women before complications occur. The application uses data from the Kenya Demographic and Health Survey 2022 and incorporates Bayesian inference for uncertainty quantification and an AI API for plain-language explanation generation.

What We Were Testing:

Your session was used to evaluate the usability, clarity, and clinical utility of the application interface, specifically whether the risk score, explanations, and referral letters are understandable and actionable for Community Health Workers without data science training.

Use of Your Feedback:

Your responses will be used to refine the application interface and explanation templates. All data is anonymised and will be reported in aggregate only. No individual responses will be identifiable.

If You Have Concerns:

If you have any questions or concerns about this study, please contact the researcher at [researcher email] or the USIU-Africa IRB at [IRB contact]. You may also contact NACOSTI at [NACOSTI contact] if you have concerns about the ethical conduct of this research.

Thank you for your contribution to improving maternal health outcomes in Kenya.

Researcher Signature: *S.T.M* Date: 17/08/2026

Appendix J: PI and Research Team Confidentiality Agreement

RESEARCH CONFIDENTIALITY AGREEMENT

Study Title: Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference

Institution: United States International University – Africa (USIU-Africa)

Principal Investigator: Selmah Tzindori Mise | ID: 669602

I, the undersigned, agree to the following conditions governing my access to and use of research data collected or analysed as part of the above-named study:

- I will not disclose any participant data, research findings, or proprietary analytical methods to any unauthorised third party.
- I will use the KDHS 2022 dataset solely for the purposes stated in the DHS Program data use agreement and the NACOSTI-approved research protocol.
- I will not attempt to re-identify any individual represented in the KDHS 2022 dataset.
- I will store all research data on password-protected devices and/or encrypted cloud storage accessible only to authorised research team members.
- I will report any accidental breach of data confidentiality immediately to the supervisor and USIU-Africa IRB.
- I understand that violation of this agreement may result in disciplinary action by USIU-Africa and revocation of DHS Program data access.
- This agreement remains in effect for the duration of the study and for five years following its completion.

Principal Investigator Signature: *S.T.M* Date: 17/08/2026

Supervisor Signature: _____ Date: _____

Appendix K: DHS Program Data Authorization Letter

The official authorization letter from the DHS Program confirming data access approval is attached on the following page.

May 28, 2026

Selmah Tzindori
United States International University
Kenya
Request Date: 05/28/2026

Dear Selmah Tzindori:

This is to confirm that you are approved to use the following Survey Datasets for your registered research project titled: "Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference":

Kenya

To access the datasets, please login at: https://www.dhsprogram.com/data/dataset_admin/login_main.cfm. The user name is the registered email address, and the password is the one selected during registration.

The IRB-approved procedures for DHS public-use datasets do not in any way allow respondents, households, or sample communities to be identified. There are no names of individuals or household addresses in the data files. The geographic identifiers only go down to the regional level (where regions are typically very large geographical areas encompassing several states/provinces). Each enumeration area (Primary Sampling Unit) has a PSU number in the data file, but the PSU numbers do not have any labels to indicate their names or locations. In surveys that collect GIS coordinates in the field, the coordinates are only for the enumeration area (EA) as a whole, and not for individual households, and the measured coordinates are randomly displaced within a large geographic area so that specific enumeration areas cannot be identified.

The DHS Data may be used only for the purpose of statistical reporting and analysis, and only for your registered research. To use the data for another purpose, a new research project must be registered. All DHS data should be treated as confidential, and no effort should be made to identify any household or individual respondent interviewed in the survey. Also, be aware that re-distribution of any DHS micro-level data, either directly or within any tool/dashboard, is not permitted. Please reference the complete terms of use at: https://dhsprogram.com/Data/terms-of-use.cfm.

The data must not be passed on to other researchers without the written consent of DHS. However, if you have coresearchers registered in your account for this research paper, you are authorized to share the data with them. All data users are required to submit an electronic copy (pdf) of any reports/publications resulting from using the DHS data files to: references@dhsprogram.com.

Sincerely,

Olsen Hanner

Olsen Hanner
Data Archivist
The Demographic and Health Surveys (DHS) Program